



Structural Barriers to Healthy Food and High-Sugar Consumption in Black Young Adult Women: The Mediating Role of Internal Health Locus of Control

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Abstract

Background Consistent overconsumption of high-sugar food and beverages (HSFB)—defined as consuming more than 25 g of added sugars per day [1]—is a health risk factor for Black young adult women in the United States. A diet high in added sugar can increase women’s likelihood of developing chronic conditions, like obesity, for which they are already at a high risk. Research often focuses on individual-level contributors to this eating behavior; however, studies infrequently consider how structural barriers may affect individuals’ eating habits.

Purpose The current study examines whether experiencing more structural barriers to accessing perceived healthy food is associated with amount of consumption of HSFB and if this association is mediated by one individual-level factor - internal health locus of control.

Methods Data come from 504 Black young adult women ($M_{age} = 24.7$, 98.4% cisgender) who completed questionnaires via an online survey.

Results Experiencing more structural barriers to accessing perceived healthy food was associated with more consumption of HSFBs ($\beta = 0.05$, $p = .013$, 95% CI = 0.01, 0.09). Specifically, the availability of fast food near where women lived, difficulties accessing transportation to purchase healthy foods, and the cost of food or grocery delivery are associated with more HSFB consumption. Internal health locus of control did not mediate the association between barriers and consumption (95% CI = -0.01, 0.02).

Conclusions More research is needed to assess the mechanisms linking structural barriers and HSFB consumption. Addressing contextual factors, like structural barriers, may be a critical step to equitably improve eating behaviors for Black young adult women.

Keywords Black · Young adult · Women · Structural barriers · Health locus of control · Diet

Approximately four out of five Black adult women in the United States have overweight (classified as a body mass index of 25.0–30.0) or obesity (body mass index of 30.0 or greater), which can increase their risk for chronic conditions such as diabetes and heart disease [2]. Early in the life-course, Black young adult women have a higher prevalence of overweight and obesity compared to Black young adult men and young adults from other racial and ethnic groups

[3]. Young adulthood is a period in which excess amounts of added sugars (i.e., over 25 g per day for women; [1]) are consumed [4]. This is inclusive of high-sugar foods and beverages such as desserts, sweet snacks, coffee, tea, and sugar-sweetened beverages [1]. Consumption of high-sugar foods and beverages can increase the risk of overweight or obesity [5]. The association between consumption of high-sugar foods and beverages and obesity is complex, entailing a combination of biological, psychological, and environmental factors [6]. Nevertheless, there is often a bias that those who engage in unhealthy eating are individually responsible for their behavior and their ability to improve is solely a function of their own volition [6].

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For Black young adult women, the consumption of high-sugar foods and beverages can increase their risk of developing or worsening conditions like obesity, heart disease, diabetes, and chronic kidney disease, all of which disproportionately affect this group [7–9]. This vulnerability is especially salient during young adulthood, a highly transitional period of life encompassing milestones like attending college, entering the workforce, establishing close relationships, and starting a family [10]. For Black young adults, this developmental stage can lead to inconsistent access to high-fiber foods, including fruits, vegetables, and whole grains, and greater consumption of prepackaged and processed foods that contain added sugars [11, 12]. Research has shown that while sugar intake has generally declined since the 2000s [13, 14], it remains elevated among Black women and young adults [15, 16]. This suggests that Black young adult women may have a greater likelihood of consuming high-sugar food and beverages. Given that young adulthood is a critical developmental period for establishing lifelong health behaviors [17, 18], it is necessary to identify factors driving high-sugar food and beverage consumption for Black young adult women to clarify areas to intervene early and prevent the early onset of chronic diseases.

Structural barriers to accessing perceived healthy food may affect Black young adult women's ability to engage in healthy eating. We define structural barriers as rooted in disproportionate inequities Black people face due to societal marginalization [19]. Structural barriers pose impediments to access and consumption of safe and nutritious food, specifically *perceived* healthy food, meaning that what is considered "safe and nutritious food" may vary by person. Structural barriers can include limited or uncertain access to adequate food (food insecurity) [20] or disproportionately fewer grocery stores in one's neighborhood but an oversaturation of fast-food restaurants and convenience stores [21]. Additionally, high costs associated with purchasing healthier food [22] or having fewer fresh food options available at the nearest grocery stores, supermarkets, or convenience stores [23] can be barriers. Lack of availability of healthy food, consumption of readily accessible nutrient-poor energy-dense foods [24], and limited knowledge, time, and financial resources are also common [25]. These barriers can all lead to unhealthy eating, which can encompass eating behaviors like high consumption of sugar-sweetened beverages and foods high in added sugar [26–30].

Black young adult women may be disproportionately affected by structural barriers to accessing perceived healthy food, which may lead them to consume high-sugar foods and beverages. However, existing literature on structural barriers and consumption of high-sugar foods and beverages is limited in two notable ways. First, many existing studies examine only one or two structural barriers to

accessing healthy food, with a preponderance of research on food insecurity [20, 21, 28, 29, 31–34]. Second, associations between structural barriers to accessing perceived healthy food and their consequences largely have been examined for Black adults writ large or older Black women, and there is a paucity of investigations specific to Black young adult women (i.e., at the intersection of race, gender, and age). Black people experience greater structural barriers to accessing healthy food compared to other racial and ethnic groups [20, 32], and research with older Black adult women suggests that they are particularly at risk for food insecurity [35, 36]. Although not specific to Black women, research with Black young adults indicates that structural barriers to accessing healthy food is a prominent issue for this age group [28, 31, 37, 38]. In one of the few studies with Black young adult women, 22% of the sample reported being food insecure [34]. In the current study, we fill gaps in the literature by examining the associations between a wider range of structural barriers and consumption of high-sugar food and beverages. We examine these associations specifically for Black young adult women, who may be at disproportionate risk of consumption of high-sugar food and beverages [5].

The food-insecurity obesity paradox states that people experiencing food insecurity, particularly women, face an elevated risk of having overweight or obesity [34, 35]. This is because individuals experiencing food insecurity may alternate between periods of hunger and consumption of nutrient-poor energy-dense foods (e.g., high-fat, high-sugar, high-calorie foods), which typically cost less than healthier options like lean meats, vegetables, and fruits [24]. According to this theory, increased energy intake may be an adaptive response to threats to food security [25]. When Black young adult women experience barriers to accessing food, specifically healthy food, this may increase their consumption of less nutritious food options. Therefore, in accordance with the food-insecurity obesity paradox, we predict that more structural barriers to accessing perceived healthy food will be associated with more consumption of high-sugar foods and beverages for Black young adult women [25, 35].

One individual-level factor that may explain the association between structural barriers to accessing perceived healthy food and consumption of high-sugar foods and beverages is internal health locus of control, which is defined as an individual's belief that they can control their health and that their health is contingent on their own actions and behaviors [39–41]. Black young adult women who experience structural barriers may especially feel that they won't succeed no matter what effort they put in to make healthy choices because of these barriers beyond their control. Although research on internal health locus of control has largely failed to consider the role of structural barriers, some research does support the hypothesis that experiencing more

barriers may reduce internal health locus of control. For instance, food insecurity is associated with less self-efficacy for healthy eating [42–44]. In turn, individuals with low internal health locus of control and the related construct of self-efficacy are less likely to engage in unhealthy behaviors [45, 46]. Lower internal health locus of control or self-efficacy has been associated with consuming a diet low in fruit and vegetables [46] and high in fat and excess sugars [45, 47], including more consumption of sugar-sweetened beverages and sugary snacks in samples of adolescents and young adults [48, 49]. However, the majority of this research is not specific to Black young adult women and largely focuses on the singular structural barrier of food insecurity. Therefore, we aim to examine whether Black young adult women who face more structural barriers to accessing perceived healthy food subsequently experience lower internal health locus of control and in turn, consume more high-sugar foods and beverages.

In the current study, we aim to examine if internal health locus of control explains the association between structural barriers to accessing perceived healthy food and consumption of high-sugar food and beverages. We hypothesized that experiencing more structural barriers to accessing perceived healthy food would be associated with more consumption of high-sugar food and beverages (i.e., total effect; Hypothesis 1).

We propose a significant indirect effect of internal health locus of control on high-sweet foods and beverages, such that the association between structural barriers to accessing perceived healthy food and consumption of high-sugar foods and beverages will be partially mediated through internal health locus of control (i.e., indirect effect; Hypothesis 2). We predicted that experiencing more structural barriers to accessing perceived healthy food would be associated with lower internal health locus of control (i.e., direct effect; Hypothesis 3) and that lower health locus of control would be associated with more consumption of high-sugar food and beverages (i.e., direct effect; Hypothesis 4).

Methods

Participants

Data come from a larger study of the health of 504 Black young adult women from the United States ($M_{age} = 24.70$, $SD_{age} = 4.90$). Only Black women (with ethnic identities across the African diaspora; see Table 1) between the ages of 18 and 35 who resided in the United States were eligible to participate. The majority of participants identified their Black ethnic background as African American (74.40%) and almost all participants identified as cisgender women (98.40%). More than half of the participants were

Table 1 Demographic characteristics of participants ($N=504$)

Variable	Mean (SD) or Median	Per-cent
Age	24.7 (4.9)	
Gender identity		
Cisgender woman		98.4%
Nonbinary		1.6%
Black ethnicity		
African American		75.2%
Caribbean Black		11.5%
Afro-Latinx		1.6%
African		6.9%
Multiracial		4.8%
Born in US		96.4%
Student status		
Not currently a student		54.2%
Yes, college student		37.1%
Yes, graduate student		8.7%
Household Income	\$45,000 - \$54,999	
Parental Education	Bachelor's degree	
COVID Past-Impact (Potential range 1–6)	1.06 (1.42)	
Region		
Northeast		16.5%
Midwest		13.5%
South		60.9%
West		9.1%

Either M (SD), Median, or % is reported, depending on type of variable. Household income and parental education are both categorical variables. To be more parsimonious, the mean levels of these variables are presented instead of being tabulated

not currently students (54.20%) and the sample's average household income was between \$45,000 and \$54,999. Less than 1% (0.60%, $n=3$) of participants reported being diagnosed with diabetes.

Procedure

The larger study (conducted from September to November 2021) was approved by Institutional Review Board at North Carolina State University. Participants were recruited from across the United States using the online research crowdsourcing platform Prolific, which has been cited as the only online behavioral research platform that provides quality participant responses high in attention, comprehension, and honesty [50]. Potential participants who meet eligibility criteria and are enrolled in Prolific's participant panel received information about the study. They then accessed the link to an online Qualtrics survey via the Prolific platform. After completing electronic informed consent via this survey, participants re-answered eligibility questions. If participants were eligible, they completed a series of questionnaires that were presented in a randomized order. In accordance with best practices [51], the survey included a mix of types of bot

screening tools to ensure data quality and participants were notified at the outset that failure on any of these screening tools would result in removal of their data from the study and no compensation. Question-based screening tools included ReCAPTCHA and two instructional manipulation check questions (i.e., question is about favorite color but asked to enter the word “table” in the text box; given information about trains and asked to select the letter of the train that goes north from a series of multiple choice options). In addition, we used data-based screening tools including (1) consistency in responding to sociodemographic questions in the survey and in a prior Prolific screener, (2) open-ended response flags for nonsensical and identical qualitative responses to five open-ended questions on the survey. Participants who failed the ReCAPTCHA ($n=2$; $<0.5\%$ of original responses), instructional manipulation check questions ($n=4$; $<0.8\%$ of original responses), or whose open-ended responses were flagged ($n=2$; $<0.5\%$ of original responses) were not included in the study and were not compensated for their participation. Participants with discrepancies between their sociodemographic Prolific screener questions and our sociodemographic screener questions ($n=1$; $<0.2\%$ of original responses) were exited from the survey before completing any of the study materials. Following completion of the survey, participants were compensated \$9.50 US per hour.

Measures

Structural Barriers to Accessing Perceived Healthy Eating

Self-reported access to perceived healthy eating was measured using nine items developed for the current study. Although there are questionnaires that examine access to healthy food, they are conducted predominantly with older adult populations. Therefore, we used the existing body of literature on structural barriers to accessing perceived healthy eating [23, 52] to develop our own measure that included items about access, especially relevant to young adults during the study collection period (e.g., use of technology for food delivery, COVID-19 precautions). Participants were asked to report the extent to which structural factors impacted their ability to eat healthy foods in the past month on a seven-point Likert-type scale from 0 (*not at all impactful*) to 6 (*extremely impactful*). It is important to note that this measure assesses participants’ reports of barriers to accessing perceived healthy eating, meaning it does not assess objective healthy eating.

Before proceeding with use, we wanted to assess whether the items followed a unidimensional factor structure, since previous literature (e.g., Caspi et al., [53]) suggests that there are five dimensions of food access. Initial items included eight pre-formulated structural factors and one open-ended

item that was recoded. These items included: (1) the cost of healthy foods, (2) the availability of healthy food near where I live, (3) the quality (e.g., freshness, variety, seasonality) of healthy foods near where I live, (4) the availability of fast food near where I live, (5) the number of options available for places to purchase healthy foods near where I live, (6) difficulties accessing transportation to get to places to purchase healthy foods, (7) concerns about the safety of going out to purchase healthy foods during COVID-19,¹ (8) the cost of food delivery or grocery delivery, and (9) something else we haven’t considered. Open-ended responses to item nine were examined and central themes were derived to either create a new item or re-code responses into a pre-existing item. This generated one new item, which was called “lack of material resources” and included structural barriers like “storage for healthy foods” and “access to a kitchen.” Since it was not feasible to collect a new sample to do an exploratory factor analysis (EFA) and confirmatory factor analysis (CFA), we used the split sample strategy (see Cudeck & Browne [54]), to conduct the EFA with a random sample of ~50% of the cases ($n=248$). Then, we conducted the CFA with the full sample. After conducting the EFA and CFA, two items were removed (see Supplemental Appendix). A total mean score was calculated from the remaining eight items such that a higher score indicates having more structural barriers to accessing perceived healthy food. The internal consistency of the measure in the current sample was acceptable (Cronbach’s $\alpha=0.84$).

Internal Health Locus of Control

The internal health locus of control subscale of the Multidimensional Health Locus of Control scale was used to assess participants’ feelings of control over their health [41]. Participants rated their agreement with six items on a six-point Likert-type scale from 1 (*strongly disagree*) to 6 (*strongly agree*). An example item is “If I get sick, it is my own behavior which determines how soon I get well again.” The total subscale score was created by averaging items, such that higher scores indicate a greater internal health locus of control. The internal consistency of the measure in the current sample was acceptable (Cronbach’s $\alpha=0.72$).

Consumption of High-Sugar Foods and Beverages

Participant’s average consumption of high-sugar food and beverages was measured using one item from the diet subscale of the Diabetes Self-Care Activities measure, which was modified for non-diabetic populations [55]. While this

¹ This item was added because the survey was conducted when many COVID-19 restrictions were still in place.

measure has not been formally validated in samples of Black women / young adult women, several studies have used this measure in predominantly Black samples [e.g., 56–60]. Participants indicated what *approximate* percentage of their meals during the past week “included sweets and desserts such as pie, cake, jelly, soft drinks (regular, not diet drinks), cookies?” on a Likert-type scale with response options including: 0 (*no high-sweet foods; 0%*), 1 (25%), 2 (50%), 3 (75%), 4 (*all of meals; 100%*). Higher scores indicated greater consumption of high-sugar food and beverages.

Covariates

Sociodemographic factors can impact Black young adult women’s access to healthy food and subsequent consumption of high-sugar food and beverages [61]. Therefore, we measured seven potential covariates. We considered age and student status because in young adults who attend university, consumption of high-fat and high-sugar foods are greater in younger young adults who attend university [62]. Participants entered a number for their age and indicated if they were currently a student using the following options: No (coded as 0) and Yes (undergraduate or graduate student; coded as 1). We also measured household income and parental education as proxies of socioeconomic status. As over half of our sample was under 24 years old and this age group is still not fully independent, often relying on their parents for financial support [63], we used a parental index of education as an additional proxy for socioeconomic status. As household income can impact diet choices [29], participants indicated their total annual household income from a range of presented amounts in \$10,000 increments from 1 (\$25,000) to 10 (Over \$100,000). We also measured parental education (i.e., highest level of education completed by parents or guardians) because it is a predictor of nutrition-related health risk in Black young adult women [64]. Participants selected their parental education from eight categories, with options ranging from “Less than high school” to “Doctoral or professional degree.” The experience of living through the COVID-19 pandemic disproportionately impacted Black communities. Specifically, the pandemic affected food access and affordability in Black communities, increasing the likelihood of high-sugar consumption [33]. Therefore, participants indicated the degree to which COVID-19 impacted their health negatively in the past month on a six-point Likert-type scale from 1 (*Not at all*) to 6 (*Extremely*).

Data Analytic Plan

A power analysis using Monte Carlo simulation for a mediation model [65] using a confidence level of 95%, 10,000 replications,

and moderate correlations between variables ($r = .20$) indicated that 96% power to detect an effect for the most complex model parameter (i.e., the indirect effect) was achieved with a sample size of 504. There were no missing data.

We first ran descriptive statistics to describe sample characteristics and used preliminary analyses to determine which of our variables needed to be included as sociodemographic covariates in the serial mediation. We examined seven potential sociodemographic covariates in our analysis (age, student status, household income, parent education, COVID-19 impact). Bivariate correlations, one-way ANOVAs, and chi-squares were run to assess which should be included in our final models. To be included as a covariate, the variable had to be significantly associated (defined as $p < .05$) with both a predictor and an outcome variable, to appropriately reduce confounding of primary statistical associations of interest [66]. The final covariates that met this criterion and were therefore included in our model were age and household income.

To answer our substantive research questions, we conducted a Poisson regression mediation model with maximum likelihood estimation using Mplus version 8.3 [67]. In these analyses, the structural barriers to accessing perceived healthy food (predictor) and internal health locus of control (mediator) were both modeled as continuous, and consumption of high-sugar foods and beverages (outcome) was modeled as a count variable. A Poisson regression was the most appropriate test considering our outcome variable type. In line with previous research, we tested for overdispersion to ensure that using a standard Poisson regression model was most appropriate versus an overdispersed Poisson regression model or negative binomial regression [68]. We first graphed the outcome variable, which appeared to be fairly normally distributed. Then, we conducted the model as a Poisson regression and as a negative binomial regression and compared the AIC values [69]. The Poisson regression model had the lowest AIC values and was therefore retained. Additionally, in the negative binomial regression model, the dispersion parameter had a coefficient of zero which indicates that the model is better estimated using Poisson regression. To assess whether each hypothesis was supported or not supported we examined p-values to see if they were below our predetermined significance value of 0.05, which would indicate that the hypothesis was supported. We also examined 95% confidence intervals to see if they did not contain zero, which would also indicate that the hypothesis was supported. To assess mediation, we used the Monte Carlo method with 20,000 repetitions. From this simulation [70], we estimated a 95% confidence interval for the indirect effect (i.e., the product of the a and b paths). The indirect effect was deemed significant if it did not include 0.

Results

Descriptive Statistics and Preliminary Analyses

Table 2 presents descriptive statistics and correlations among the study variables. On average, participants reported that between 25% and 50% of their meals contained high-sugar food and beverages within the past week. Women in our sample were moderately impacted by structural barriers to perceived healthy eating ($M=3.33$, $SD=1.64$) and reported fairly high levels of internal health locus of control ($M=4.37$, $SD=0.80$).

Mediation

Table 3 reports the direct, indirect, and total effects of the mediation paths in the model. Approximately 2% of the variance in consumption of high-sugar food and beverages was explained by structural barriers to perceived healthy eating, internal health locus of control, and covariates (Pseudo $R^2 = 0.02$). The confidence interval for the model indirect effect included zero (95% CI = -0.01 , 0.002), indicating that mediation was not supported.

Hypothesis 1

In support of hypothesis 1, the association between structural barriers to accessing perceived healthy food on consumption of high-sugar food and beverages was significant. For every 1-unit increase in structural barriers to perceived healthy eating, the log of the expected count of high-sugar foods and beverage consumption increased by 0.05 units ($\beta=0.05$, $p=.013$, 95% CI= 0.01 , 0.09). In other words, experiencing more structural barriers to perceived healthy

Table 2 Descriptive statistics and intercorrelations among study variables

	1	2	3	4	5
1. Age	--				
2. Household Income	0.04	--			
3. Internal Health Locus of Control	0.11*	0.02	--		
4. Structural Barriers to Accessing Perceived Healthy Food	-0.12**	-0.19**	0.07	--	
5. High-sugar Food Consumption	-0.14**	-0.13**	-0.02	0.15**	--
M	24.72	3.57	4.37	3.48	1.36
SD	4.89	2.85	0.80	1.79	1.00
Range	18–35	0–8	1–6	0–7	0–4

Mean (M), standard deviation (SD), and range only presented for continuous and count variables. Range indicates the minimum and maximum potential values for scale scores and the observed values for age. * $p<.05$. ** $p<.01$

Table 3 Direct, indirect, and total effects of structural barriers to accessing perceived healthy foods on high-sugar food and beverage consumption through internal health locus of control ($N=504$)

Direct effects	β	95% CI	IRR	R^2
Outcome: IHLC				0.02
Age	0.12	[0.04, 0.19]**	1.04	
SBHF	0.08	[0.01, 0.16]	1.02	
Outcome: HSFB				0.01
Age	-0.09	[-0.15, -0.04]**	0.98	
Household income	-0.08	[-0.13, -0.02]*	0.97	
SBHF	0.09	[0.03, 0.14]*	1.05	
IHLC	-0.08	[-0.07, -0.05]	0.99	
Indirect effect				
SBHF $\rightarrow$ IHLC $\rightarrow$ HSFB	-0.01	[-0.01, 0.01]	–	
Total effect				
Outcome: HSFB				
SBHF	0.51	[0.23, 0.77]*	–	

β = estimated Poisson regression coefficients; IRR=Incidence Rate Ratio; IHLC= Internal Health Locus of Control; SBHF= Structural barriers to accessing perceived healthy foods; HSFB = high-sugar food and beverages consumption. * $p<.05$. ** $p<.01$

food is associated with a significantly greater proportion of consumption of high-sugar foods and beverages.

Hypothesis 2

Contrary to hypothesis 2, the associations between structural barriers to accessing perceived healthy food and internal health locus of control ($\beta=0.04$, $p=.070$, 95% CI = -0.003 , 0.08) and between internal health locus of control and consumption of high-sugar food and beverages ($\beta = -0.02$, $p=.742$, 95% CI = -0.10 , 0.07) were each nonsignificant.

Sensitivity analysis

To get more detailed insight into the specific structural barriers that led to increased high-sugar food and beverage consumption, a Spearman's correlation was conducted between each of the structural barrier items and the high-sugar food and beverage outcome. Results revealed a positive weak correlation between high-sugar food and beverage consumption and the availability of fast food near where participants lived ($r_s=0.19$, $p<.001$), difficulties accessing transportation to purchase healthy foods ($r_s=0.15$, $p=.001$), and the cost of food delivery or grocery delivery ($r_s=0.13$, $p=.003$).

Discussion

The central contributions of the current study include: examination of the association between structural barriers to accessing perceived healthy foods in consumption of high-sugar foods and beverages, and testing one

potential process in this association, specifically for Black young adult women. In support of our first hypothesis, Black young adult women who faced more structural barriers to accessing perceived healthy food reported greater high-sugar food and beverage consumption. Sensitivity analyses revealed that in this sample, the structural barriers experienced that led to increased high-sugar food and beverage consumption were availability of fast food near where women lived, difficulty accessing transportation to purchase healthy food, and the cost of food and grocery delivery. This aligns with work showing that facing structural disadvantage—such as living in a food swamp or food apartheid [35], experiencing difficulty affording food [23, 26, 71], and finding transportation to purchase healthy food [30, 72]—is associated with limited access to healthier food options. Having to navigate such barriers may lead Black young adult women to select convenient, time-saving, and cost-effective foods [26], many of which are more processed and higher in added sugar [73]. When considering the life demands during young adulthood (e.g., school, work, caretaking, etc.), selecting the more convenient and cost-effective option is less stressful than trying to purchase healthier options that may exceed what is feasible financially and timewise [26]. Another reason that experiencing greater structural barriers may be associated with greater high-sugar food and beverage consumption is that Black young adult women may use high-sugar foods to cope with the psychosocial stress of these barriers. Stress eating is the consumption of energy-rich foods high in sugar and fat in response to negative stimuli or affect [74], and is a common coping mechanism in which Black young adult women engage when experiencing stress [75–77]. Taken together, our findings indicate that structural barriers to access to perceived healthy food may be associated with the nutritional choices of Black young adult women, even among those who only faced a relatively moderate number of barriers. Our results suggest that these barriers may be associated with less healthy eating even among those who do not experience significant barriers like food insecurity.

Contrary to our second hypothesis, internal health locus of control did not explain the association between structural barriers to accessing perceived healthy food and high-sugar food and beverage consumption. Health locus of control theory would suggest that external factors like structural barriers would impact one's internal sense of control [78], however, a paucity of empirical studies have examined this proposition, and even fewer have considered multiple structural barriers for Black young adult women. It may be that Black women use adaptive and proactive coping strategies to combat any reduction in sense of control in the face of structural disadvantages. Adaptive coping strategies are generally action-oriented and include

thoughts and behaviors that aim to resolve stressors [79], and proactive coping strategies are those that are leveraged in anticipation of future stressors [80]. For Black young adult women, factors such as the superwoman schema, awareness of inequity in society, and leveraging social support may be ways they proactively cope with barriers and do so in an adaptive way for them. Superwoman schema includes a perceived obligation to project strength and succeed despite limited resources [81]. Some Black young adult women may recognize structural barriers they face but may take on a superwoman role, reframing these barriers to emphasize their own strength and focus on creating their own success to cope with the circumstances and confront life adversity [82]. When Black young adult women expect that they will experience greater structural barriers rooted in inequities in society, perhaps the mere presence of these barriers in their life may not limit their own sense of control because they do not internalize these as individual shortcomings but instead recognize them as societal flaws.

Another possible explanation is that Black young adult women facing barriers to healthy food access may also receive some degree of instrumental social support from family, friends, and community members to overcome these barriers. If they can successfully navigate any individual barriers they face, their own sense of internal health locus of control may not be affected. An important consideration is that women reported fairly low levels of impact from COVID-19, which was assessed as a covariate in this investigation. Because some of the structural barriers were tailored to COVID-19, this may indicate that they did not feel like COVID-19 imposed additional barriers. Moreover, our sample reported low levels of perceived structural barriers altogether, which suggests that they may not have perceived salient structural barriers to accessing healthy food, in general. Therefore, internal health locus of control may not be a reason why barriers increase high-sugar food and beverage consumption. It may be that if Black young adult women do not perceive these barriers as particularly salient, internal health locus of control naturally plays less of a role in guiding their consumption of high-sugar foods and beverages. When barriers are experienced (e.g., the cost of purchasing healthy food or food delivery) women may feel that these external constraints inevitably dictate what they can access and consume, thereby mitigating the influence of their own internal sense of control. It should be noted that the predictors in the model explained a small amount of variance in the outcome, suggesting that future research should continue assessing various structural barriers to accessing healthy food, specifically relevant to Black young adult women, that may affect consumption of high-sugar foods and beverages.

Limitations and Future Directions

Several limitations and constraints on the generalizability of the current study should be noted. As data are from a cross-sectional correlational study, causal claims cannot be made. The use of mediation with cross-sectional data is not ideal. However, scholars suggest that the use of mediation is appropriate for theory-driven foundational work for which associations are not currently known and more information is needed before moving to a more costly and time-consuming experimental or longitudinal design [83]. The current investigation is an example of this foundational work, which is necessary basic research for early-phase intervention development and testing. Nearly all Black young adult women in our study were cisgender and Black women with other marginalized gender identities (e.g., non-binary, transgender) may face additional structural barriers to healthy eating, so these results are not expected to replicate beyond cisgender Black women. We utilized a non-probability national sample and participants were recruited online. Although the majority of young adults are online, the results may not generalize to all Black young adults across the United States, and some who may face greater barriers to access and health may not be fully represented in our sample. The majority of participants lived in the South, but we do not have data on the degree of urbanicity of the places where they resided. Urbanicity may impact structural barriers to healthy eating, for instance, the degree of access to certain grocery stores or the time it takes to secure healthy food. Therefore, we cannot guarantee that the results of this study will replicate across different regions or would not further differ by urbanicity.

Some limitations arise from the measurement of structural barriers and high-sugar foods and beverages. A novel measure of structural barriers was used because of a lack of preexisting measures that were both sufficiently short and applicable to assess this construct in our target population. However, this measure didn't capture the length of time participants had experienced each structural barrier or the stressfulness of each barrier. Therefore, future work should consider testing the moderating role of barrier duration and stress and how this may illuminate further nuance in the role of internal health locus of control. Next, the measure of high-sugar food and beverage consumption was a single-item indicator of the construct and only asked about consumption during meals, rather than between meals (e.g., as snacks). It may be that we are missing a primary eating context in which consumption of high-sugar food and beverages occurs. Data from NHANES 2015–2018 reveals that among US adults (≥ 20 years) mean calorie intake was highest during snacking (e.g., desserts, sweets, salty snacks, sugar-sweetened beverages) among high added sugar

consumers, which corresponded to 43% of calories from added sugars [4]. This is followed by dinner (23.5%), lunch (18.7%), and then breakfast (15%). Since our item focuses on consumption across meals, our estimate of high-sugar food and beverage consumption may be more conservative as it is not disaggregated by meals and does not explicitly focus on snacking where the bulk of consumption is. Additionally, our measure relies on participant's interpretation of what constitutes high-sugar (e.g., just beverages or added sugars during cooking) and meals (e.g., all three meals or just one meal). This may mean that participants likely reported based on the most salient sources or occasions of their high-sugar food and beverage consumption.

However, by using a measure that captures the quantity of consumption, which is expressed in terms of a percentage of meals rather than in grams or other nutritional intake values, this allows participants to more easily recall their approximate intake. Additionally, as this item assesses consumption over seven days, it gives a more general sense of high-sugar food and beverage consumption rather than over just 24-hours which could be unrepresentative of participant's typical consumption patterns. This measure was the only one available in the larger study and was selected because it offered a time-efficient way to get a general sense of nutrition. Given that we examined our research questions among a non-clinical sample, this measure provides a useful baseline of meal-time consumption and can be used as a starting point for further nutrition research. Future investigations should prioritize measures that rely less on long-term recall and are less prone to potential bias such as daily diary methods or leveraging the use of biomarkers like urinary sucrose [84] and isotope ratios [85]. Studies should also focus on providing a more granular understanding of added sugar consumption separately for foods, beverages, meals, and snacks. This will be critical to accurately assess the added sugar consumption in Black young adult women.

The current investigation did not measure food insecurity in the current study and therefore was not able to account for this as a covariate. Experiencing food insecurity may have also been associated with women's consumption of high-sugar foods and beverages and could have magnified their experience with the structural barriers. Therefore, future research should include food insecurity to assess how many women were food insecure and control for this when assessing the extent to which structural barriers affect consumption of high-sugar food and beverage consumption.

Implications

Black women are disproportionately at risk for negative health consequences associated with unhealthy eating [2]

and young adults consume high amounts of high-sugar food and beverages [5]. Yet there is a paucity of research on structural barriers, internal health locus of control, and high-sugar foods and beverages at the intersection of race, gender, and age, despite the necessity of this specificity for tailored intervention efforts. This study adds to the body of literature examining how structural factors may contribute to health behaviors for Black young adult women. Our results suggest that structural-level factors are associated with Black women's consumption behaviors even relatively early in the life-course during young adulthood. Researchers can use this information to examine the environmental, psychological, and biological contexts in which structural barriers during young adulthood do and do not translate into health risk for Black women later in life. Focusing solely on individual-level attitudes necessary for behavior change may not go far enough to engender health equity for this population. As a first step, behavioral medicine practitioners could employ screenings for structural barriers to healthy food access alongside nutrition assessments for Black young adult women. New behavioral medicine interventions can be designed or existing interventions modified to account for or address structural barriers, which may require interdisciplinary collaboration (e.g., implementation science). Policy at local and national levels should be enacted to focus on mitigating these structural barriers, and ensure that they are being ameliorated equitably for disproportionately impacted populations. Taken together, identifying structural barriers to health behaviors for marginalized racial, ethnic, and gender populations remains an important priority for a behavioral medicine grounded in health equity.

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Author Contributions Abbey N. Collins: (Conceptualization [Lead]; Data Curation [Lead]; Formal Analysis [Lead]; Methodology [Equal]; Software [Lead]; Visualization [Lead]; Writing-original draft [Lead])

Data Availability De-identified data may be made available for research purposes by contacting the first author via email and completing a data use agreement.

Declarations

Ethics Approval All procedures performed to collect data for this study were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The larger study was approved by Institutional Review Board at North Carolina State University. Since the investigation is a secondary data analysis, additional IRB approval was not needed.

Consent to Participate Informed consent was obtained electronically from all individual participants included in the study.

Conflict of interest The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Transparency Statements Neither the analysis plan or the study were formally preregistered. De-identified data from this study are not available in a public archive. De-identified data from this study will be made available (as allowable according to institutional IRB standards) by emailing the corresponding author. Analytic code used to conduct the analyses presented in this study are not available in a public archive. They may be available by emailing the corresponding author. Materials used to conduct the study are not publicly available.

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